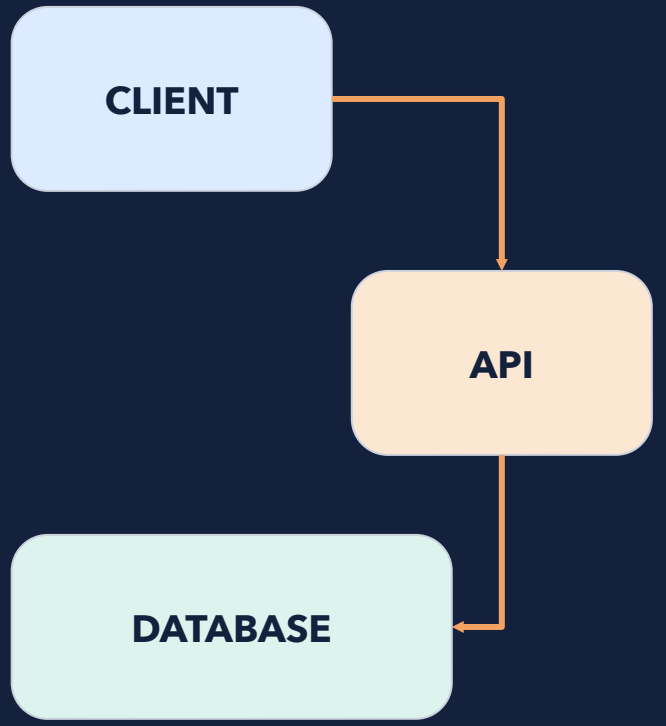
CST 4714 · WEEK 1

How Applications Use Databases

A saved request disappears from a dashboard. The stored rows explain why.



THE COURSE CASE

A repair request missing from the dashboard

One fictional repair desk will connect the application, the stored data, and the query.

Stored facts

Each ticket has an ID, a subject, a status, and an assignee.

One row represents one request.

Dashboard promise

Every active request should appear.

In this case, active means status new or open.

Reported problem

A student has confirmation 1004 for a broken bench.

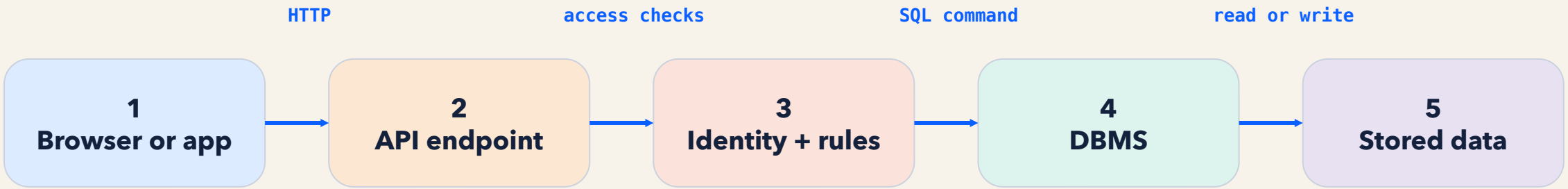
Staff cannot find it on the dashboard.

A stored record and a query result can contain different rows.

REQUEST PATH

Creating a support request

The application checks the request before asking the DBMS to store a row.



REQUEST

```
POST /api/tickets
{
  "subject": "Broken bench slat"
}
```

ILLUSTRATIVE SQL

```
INSERT INTO tickets
  (ticket_id, subject,
   status, assignee_id)
VALUES (1004, 'Broken bench slat',
        'new', NULL);
```

CORE VOCABULARY

Database, DBMS, and managed platform

Separating the layers makes architecture and troubleshooting clearer.

Database

The organized data and its structure.

Examples:
- tables, rows, and constraints
- collections and documents
- indexes and metadata

DBMS

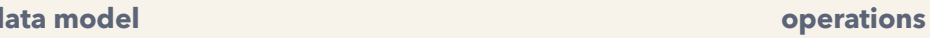
Software that manages data safely.

Examples:
- PostgreSQL
- MongoDB
- SQLite

Managed platform

A service that operates DBMS infrastructure.

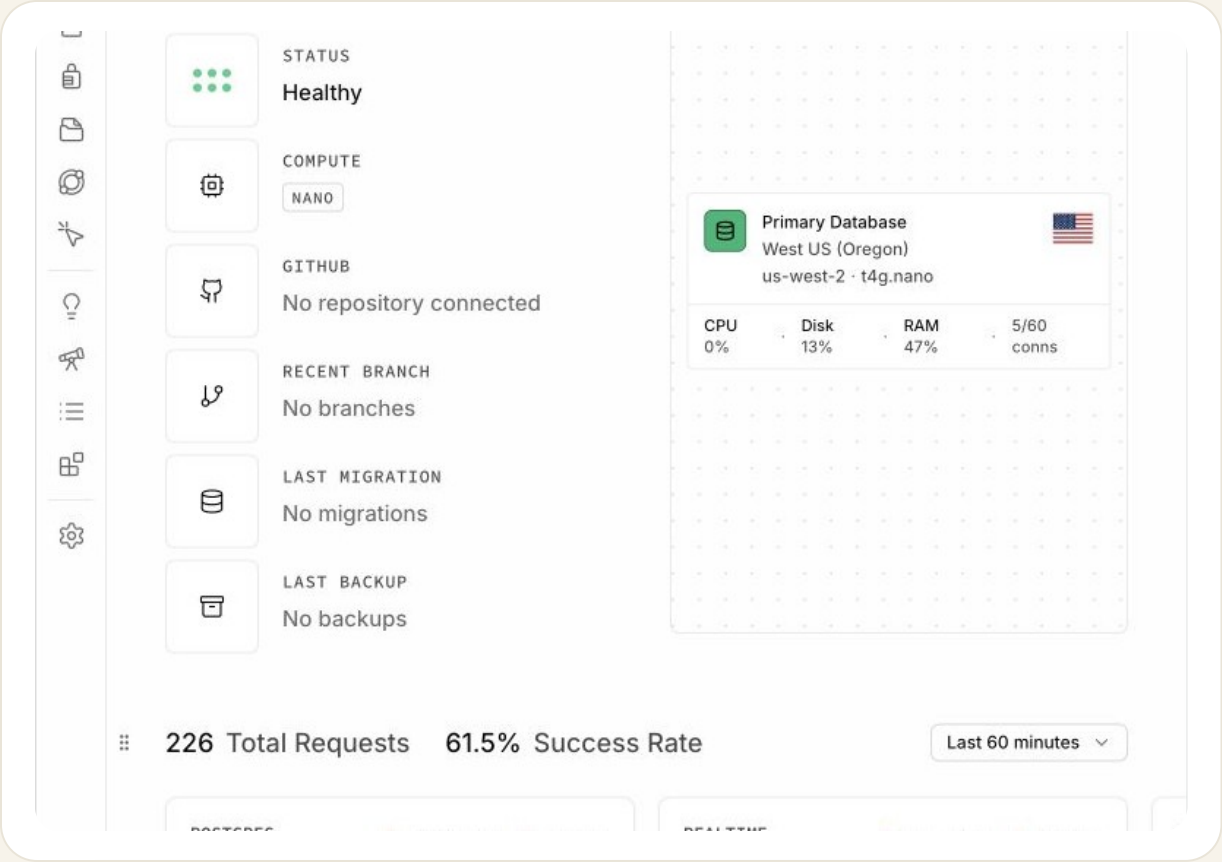
Examples:
- Supabase
- MongoDB Atlas
- Amazon RDS



CLOUD INTERFACE

Supabase is a platform built around PostgreSQL

The dashboard exposes projects and services; PostgreSQL remains the database engine.



What the platform adds

Dashboard · managed infrastructure · API services · authentication · logs · configuration

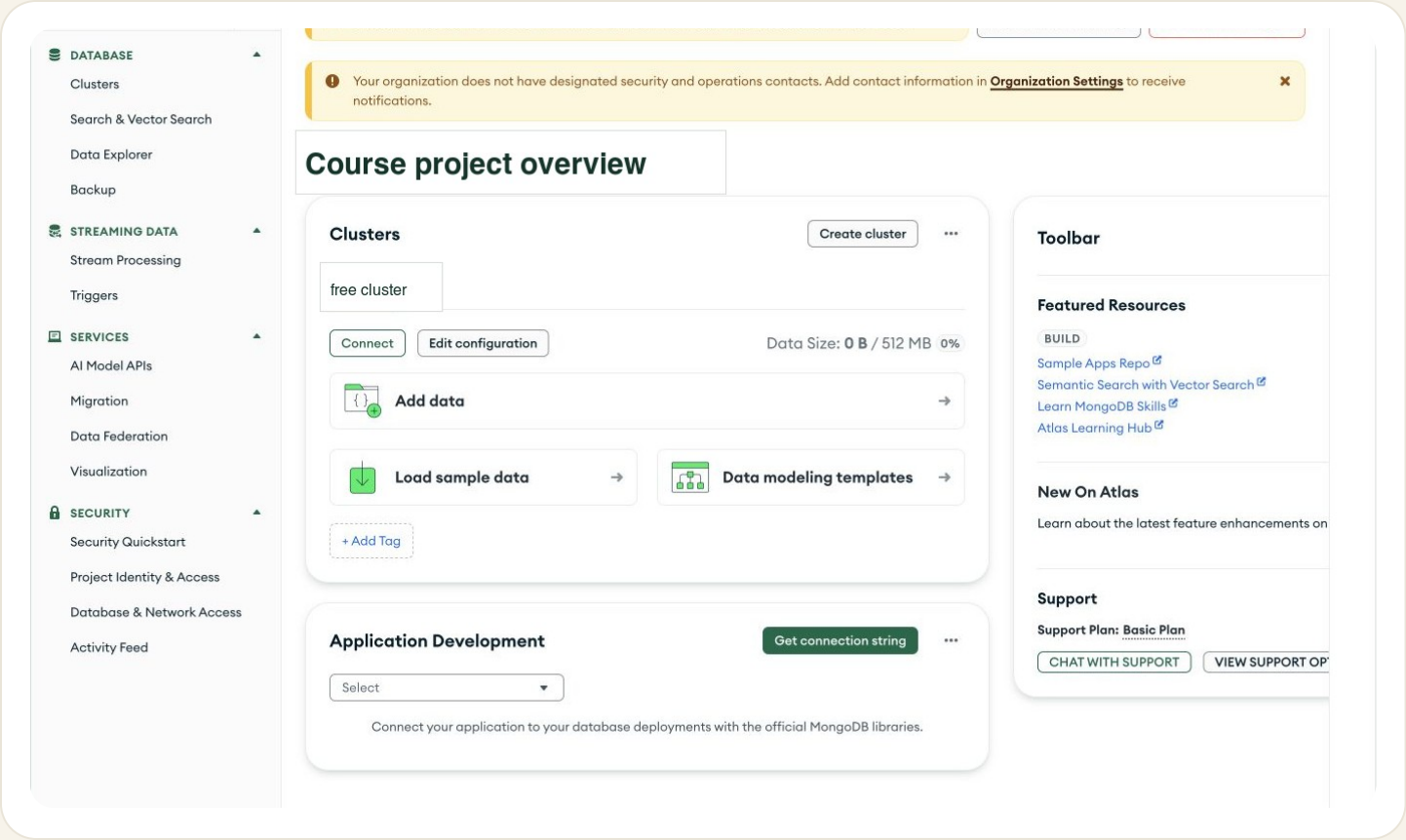
What PostgreSQL still does

Tables · SQL · constraints · transactions · query planning · indexes · roles

CLOUD INTERFACE

MongoDB Atlas operates MongoDB in the cloud

Projects, clusters, users, network rules, and data tools surround the document database.



Atlas layer

Project · cluster · network access · database users · logs

MongoDB layer

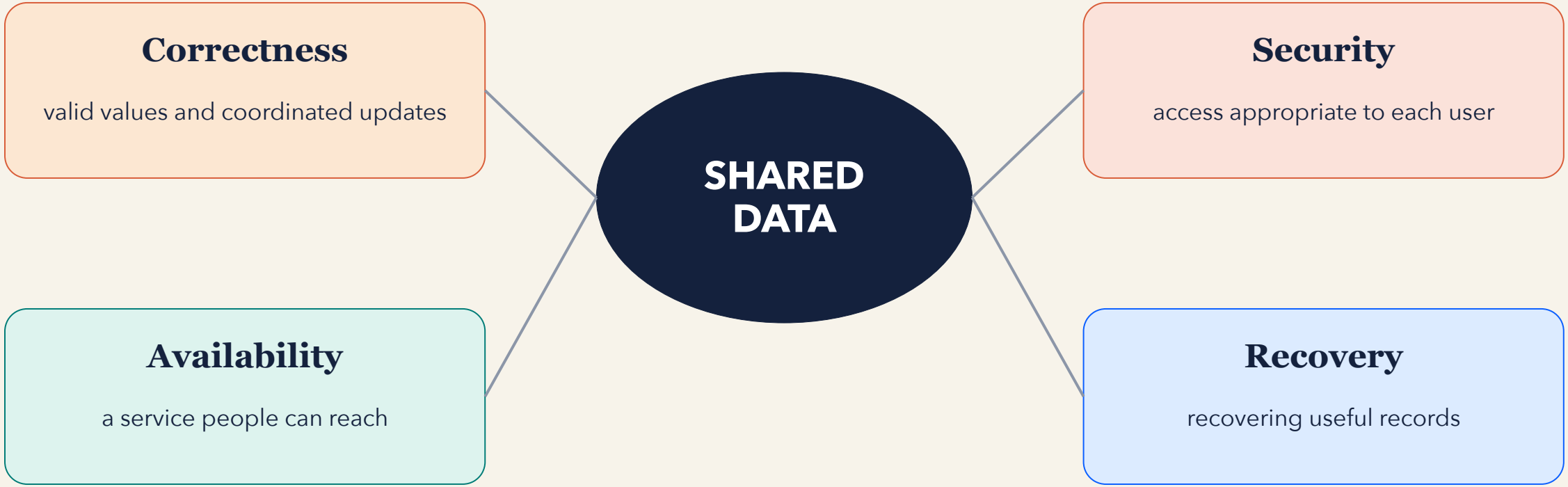
Databases · collections · BSON documents · queries · indexes

Dated interface example. No cloud account is needed for this week's lab.

TECHNICAL WORK

Responsibilities of database administration

The job is to keep shared data correct, available, secure, and recoverable.



DIAGNOSIS

A saved ticket missing from a report

The write path and the later read path need separate checks.

Observed symptom

The form reports success, but ticket 1004 is absent from the staff dashboard.

ONE OBSERVATION

```
HTTP 201 Created
{ "ticket_id": 1004,
  "status": "new" }
```

1. **Client**
   Did the browser send the request?
2. **API**
   What status and body came back?
3. **Database**
   Did the command commit?
4. **Read path**
   Is the later query filtering it out?

DAY 2

The Relational Model

A precise language for tables, questions, and transformations

A row represents one fact.
Queries produce new results.

RELATIONAL ALGEBRA

```
σ status = 'open' (tickets)
π ticket_id, subject (tickets)
```

RELATION

One row represents one support request

Three rows from the lab, with four attributes. Subjects are shortened for display.

ticket_id	subject	status	assignee_id
1001	Streetlight dark	open	201
1003	Low water pressure	resolved	201
1004	Broken bench slat	new	NULL

schema attribute names + domains

tuple one complete row of values

key stable identity

VOCABULARY

Schema and current data

The schema describes allowed structure. The instance contains the rows present now.

Schema

```
tickets(
  ticket_id integer,
  subject text,
  status text,
  assignee_id integer
)

Schema notation, not a SQL command.
```

Instance

The tuples stored at one moment.

At 10:00: 3 tickets
At 10:05: 4 tickets

Changes whenever rows are inserted, updated, or deleted.

Status values

```
new, open, resolved, closed
```

A domain constrains values. Nullability separately controls whether a value is required.

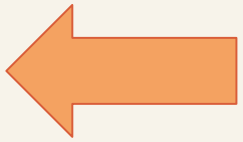
KEYS

An assignee ID refers to an agent record

A nullable foreign key permits an unassigned ticket without inventing an agent.

agents

agent_id	name
201	Priya Shah
202	Noah Williams



PK ← FK

tickets

ticket_id	subject	assignee_id
1001	Streetlight dark	201
1003	Low water pressure	201
1004	Broken bench slat	NULL

Ticket 1001 refers to agent 201. Ticket 1004 has no assigned agent yet.

RELATIONAL ALGEBRA

Selection filters rows

σ is sigma. This open-only condition keeps ticket 1001 in the displayed excerpt.

ticket_id	subject	status
1001	Streetlight dark	open
1003	Low water pressure	resolved
1004	Broken bench slat	new

σ

ALGEBRA

```
σ status = 'open' (tickets)
```

SQL

```
SELECT *
FROM tickets
WHERE status = 'open';
```

RELATIONAL ALGEBRA

Projection keeps selected attributes

π is pi. The SQL output list chooses columns. DISTINCT explicitly removes duplicates.

ticket_id	subject	status	assignee_id
1001	Streetlight dark	open	201
1003	Low water pressure	resolved	201
1004	Broken bench slat	new	NULL

π

ALGEBRA

```
π ticket_id, subject (tickets)
```

SQL

```
SELECT ticket_id, subject
FROM tickets;
```

Output: ticket_id and subject

RELATIONAL ALGEBRA

Products and joins

The full lab has four tickets and two agents. Matching IDs determine the join result.

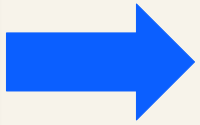
Cartesian product $R \times S$

Pairs every ticket with every agent.

|tickets| = 4 and |agents| = 2

|tickets × agents| = 8

These include unrelated pairs.



`condition`

Equi-join $R \bowtie S$

An inner join keeps matching pairs.

tickets.assignee_id
= agents.agent_id

A ticket with no matching agent produces no inner-join row.

A left join can preserve a ticket even when its assignee has no match.

TRANSLATION

The current dashboard query

Active status and a matching agent are separate requirements in this query.

MEANING

Input: tickets and agents

Match the assignee ID to an agent ID.
Keep status new or open.
Return ticket ID, subject, and name.

An unassigned ticket has no match.



SQL

```
SELECT t.ticket_id, t.subject, a.name
FROM tickets AS t
JOIN agents AS a
  ON t.assignee_id = a.agent_id
WHERE t.status IN ('new', 'open');
```

1 match agent IDs

2 keep active status

3 choose output columns

WEEK 1 RECAP

Stored data and displayed results

Next week: SQL filters, NULL, joins, summaries, and safe practice changes.

Request path

client → API → identity → DBMS → data

System layers

database ≠ DBMS ≠ managed platform

Row meaning

relations, attributes, tuples, domains, keys

Query operations

σ selection · π projection · ⋈ join

Before Week 2: review filtering, NULL, and joins. Reading links are in the weekly guide.